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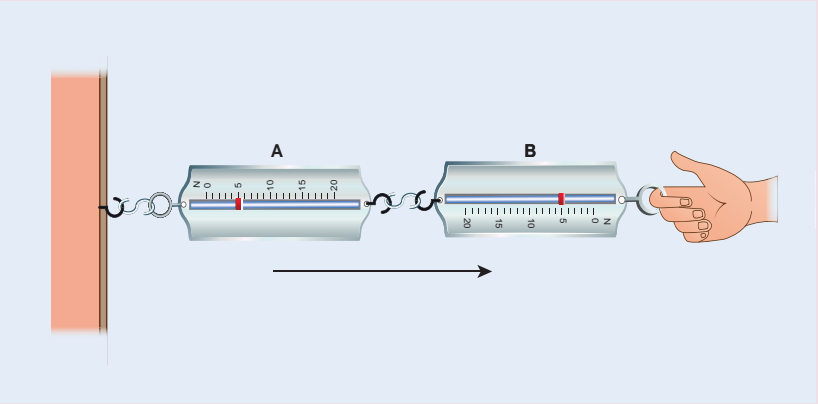
ACTIVITY

Verification of Newton's third law

Attach two spring balances as shown in the figure. Fix one end with rigid support and leave the other end free, which can be pulled with the hand.

Pull one end with some force and note the reading on both the balances.

Repeat the exercise a number of times.



Note

The reading in the spring balance A is due to the force given by spring balance B. The reading in the spring balance B is due to the reaction force given by spring balance A. Note that according to Newton's third law, both readings (force) are equal.

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3.2.4 Discussion on Newton's Laws

1. Newton's laws are vector laws. The equation $\vec{F} = m\vec{a}$ is a vector equation and essentially it is equal to three scalar equations. In Cartesian coordinates, this equation can be written as $F_x\hat{i} + F_y\hat{j} + F_z\hat{k} = ma_x\hat{i} + ma_y\hat{j} + ma_z\hat{k}$. By comparing both sides, the three scalar equations are

   $F_x = ma_x$ The acceleration along the x direction depends only on the component of force acting along the x-direction.

   $F_y = ma_y$ The acceleration along the y direction depends only on the component of force acting along the y-direction.

   $F_z = ma_z$ The acceleration along the z direction depends only on the component of force acting along the z-direction.

   From the above equations, we can infer that the force acting along y direction cannot alter the acceleration along x direction. In the same way, F_z cannot affect a_y and a_x. This understanding is essential for solving problems.

2. The acceleration experienced by the body at time t depends on the force which acts on the body at that instant of time. It does not depend on the force which acted on the body before the time t. This can be expressed as